Meteoroid Environment Office

NASA's Lunar Impact Monitoring Program

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Jack Schmitt/Apollo 17 observation of lunar impact

"NASA Apollo 17 transcript" discussion is given below (before descent to lunar surface):
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03 15 38 09 (mission elapsed time)
(10 Dec 1972, 21:16:09 UT – possible Geminid)

LMP Hey, I just saw a flash on the lunar surface!

CC Oh, yes?

LMP It was just out there north of Grimaldi [mare]. Just north of Grimaldi. You might see if you got anything on your seismometers, although a small impact probably would give a fair amount of visible light.

CC Okay. We'll check.

LMP It was a bright little flash right out there near that crater. See the [sharp rimed] crater right at the [north] edge of [the] Grimaldi [mare]? Then there is another one [i.e., sharp rimed crater] [directly] north of it [about 50km]- fairly sharp one north of it. [That] is where there was just a thin streak [pin prick] [flash?] of light.

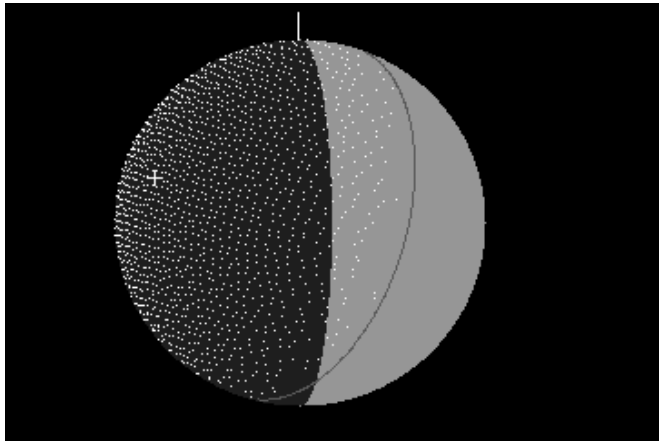
CC How about putting an X on the map where you saw it?

LMP I keep looking for -- yes, we will. I was planning on looking for those kind of things....





Jack Schmitt - NASA



Geminids 12/13/1972

Historical overview

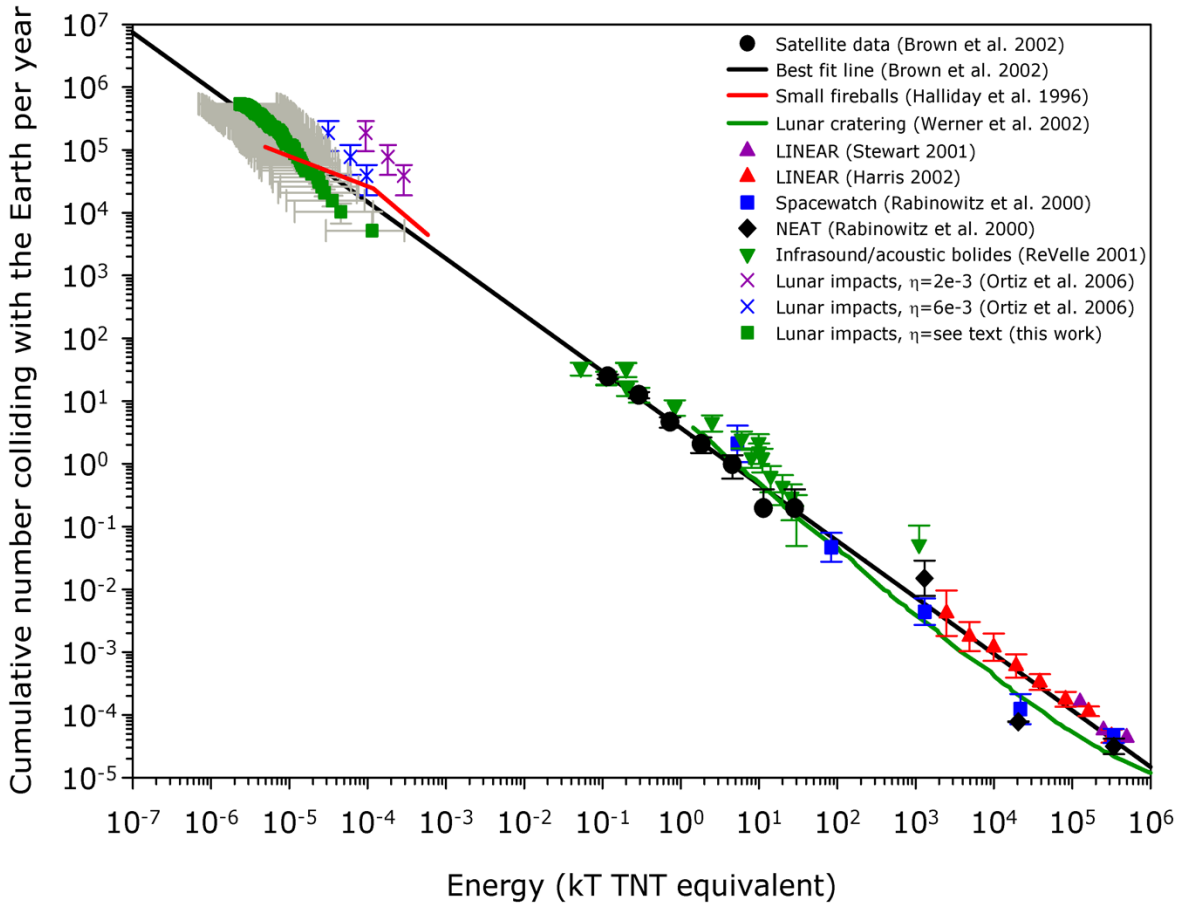
- Amateurs reported seeing impact flashes during the Leonid meteor storms of 1998-2002.
- The NASA lunar impact monitoring program began routine operations in 2006 to collect data needed for lunar impact ejecta environment definition for the old Constellation Program
- Initial 5 years were anytime the weather was good enough to permit lunar observations
  - Observations in recent years have been during showers only



Our 1st flashes were detected with this amateur telescope!

Orion Atlas 10 EQ – 10" Newtonian

- A 2014 paper published in Icarus reported the flux based on the first 5 years of observations and 126 calibrated flashes (267 hours of observation time) : Suggs et al., Icarus 238 (2014) 23-36 http://dx.doi.org/10.1016/j.icarus.2014.04.032
- The flux we determined confirmed the Brown et al. size distribution which is incorporated into the Lunar Meteoroid Ejecta Engineering Model (LMEEM) used for HLS



William Cooke – NASA MEO

Observational setup



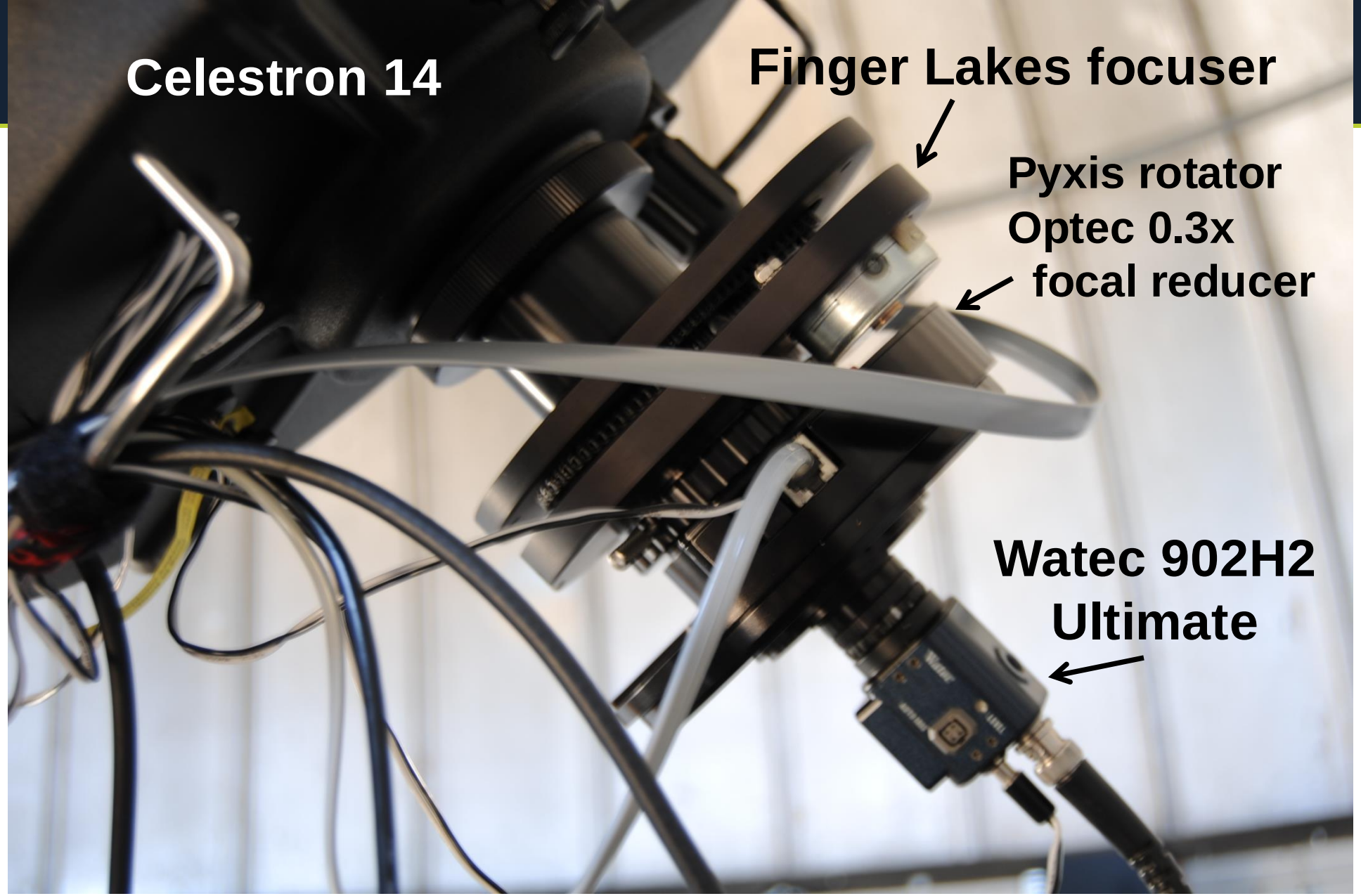
Automated Lunar and Meteor Observatory (Minor Planet Center H58)

Telescopes

- 20” RCOS (0.5m) on Paramount ME
- 14” Celestron (0.35m) on Paramount MX

Detectors

- Watec 902H2 Ultimate
- Sony HADView EX CCD
- Gamma=0.45, manual gain, autoshutter off
- 640x480 pixels


Celestron 14
Finger Lakes focuser
Pyxis rotator
Optec 0.3x
focal reducer
Watec 902H2
Ultimate

Operator position



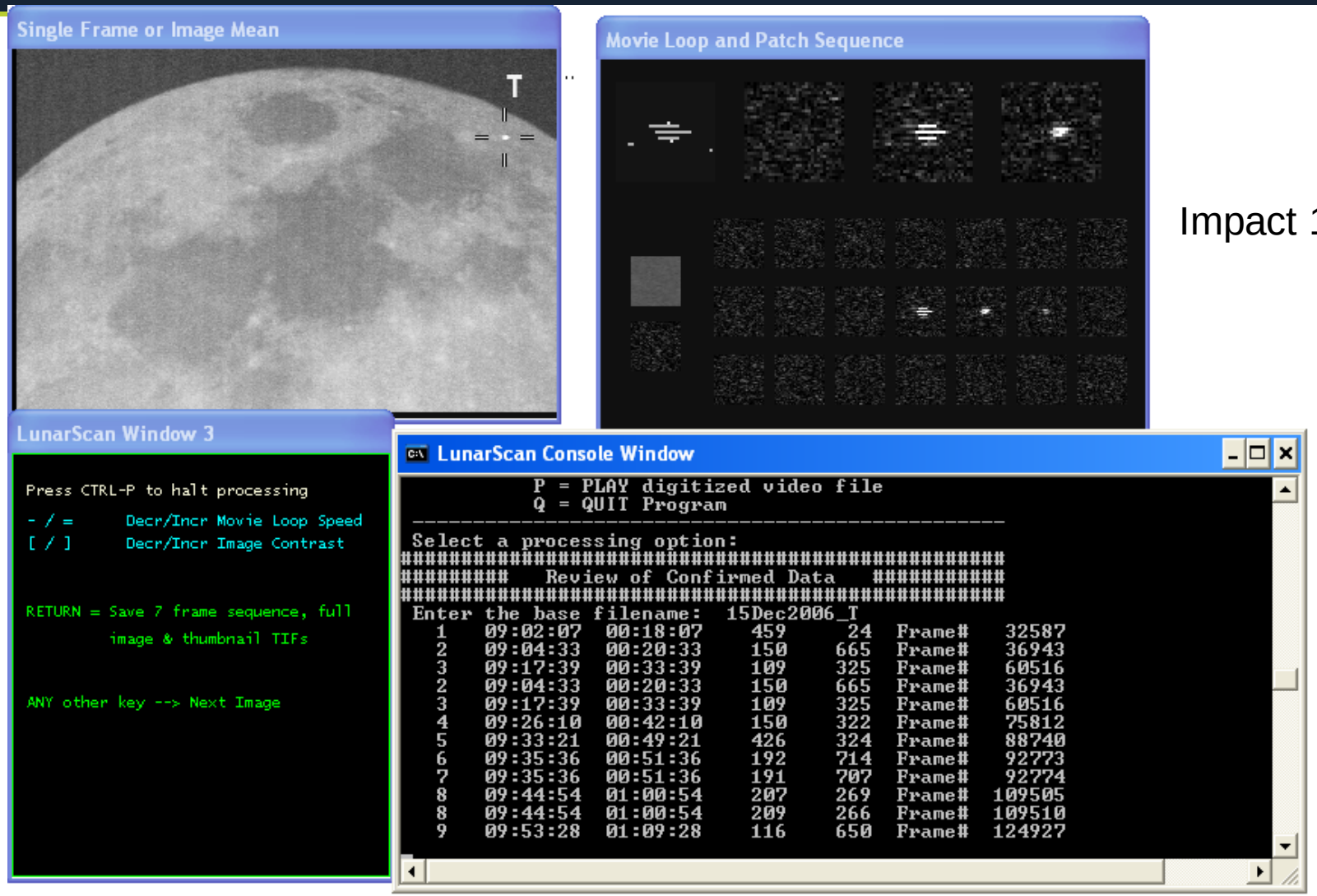

Real Time

Camera Field of View and Tracking

- Approximately 20 arcmin horizontal
- Approximately 1m effective focal length with ½ inch CCD
- Good compromise between collecting area and glare
- Telescope mount with lunar rate (in RA and Dec) is helpful although manual corrections are needed
- Aristarchus and Proclus are easy to see and use as tracking targets

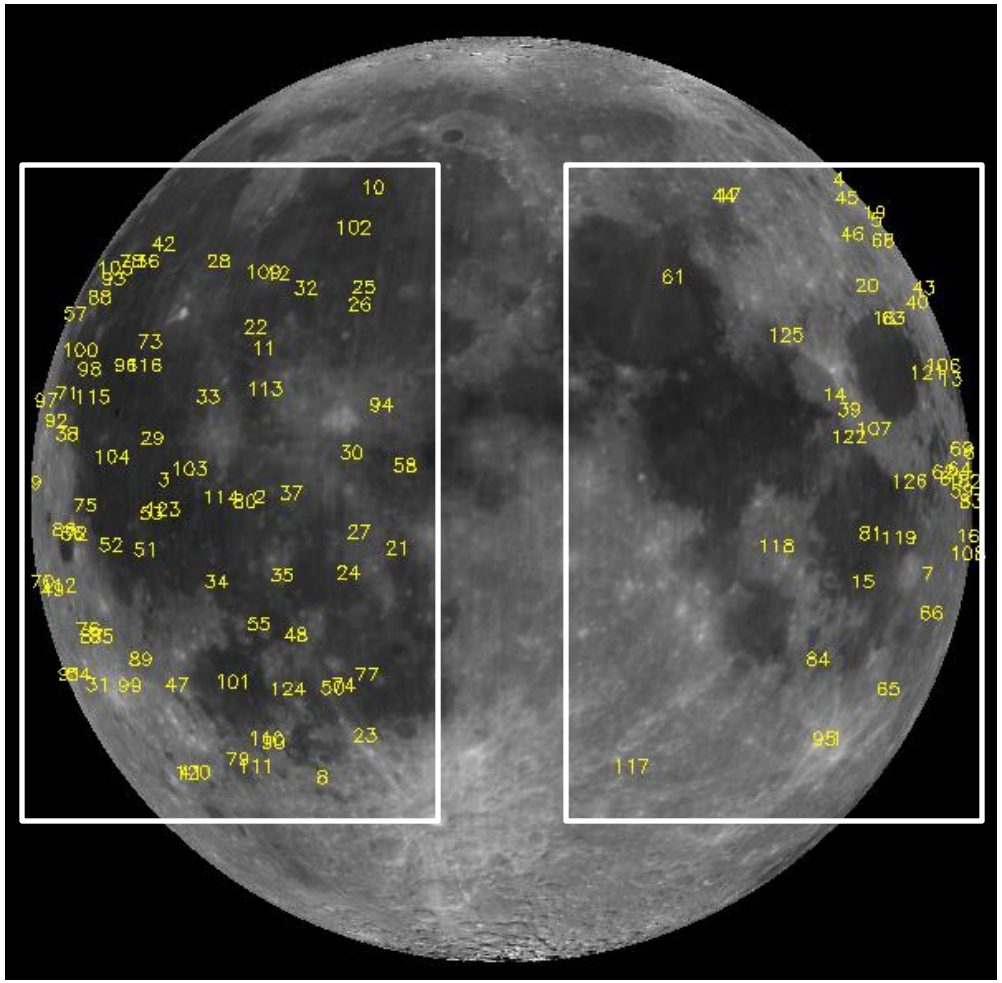


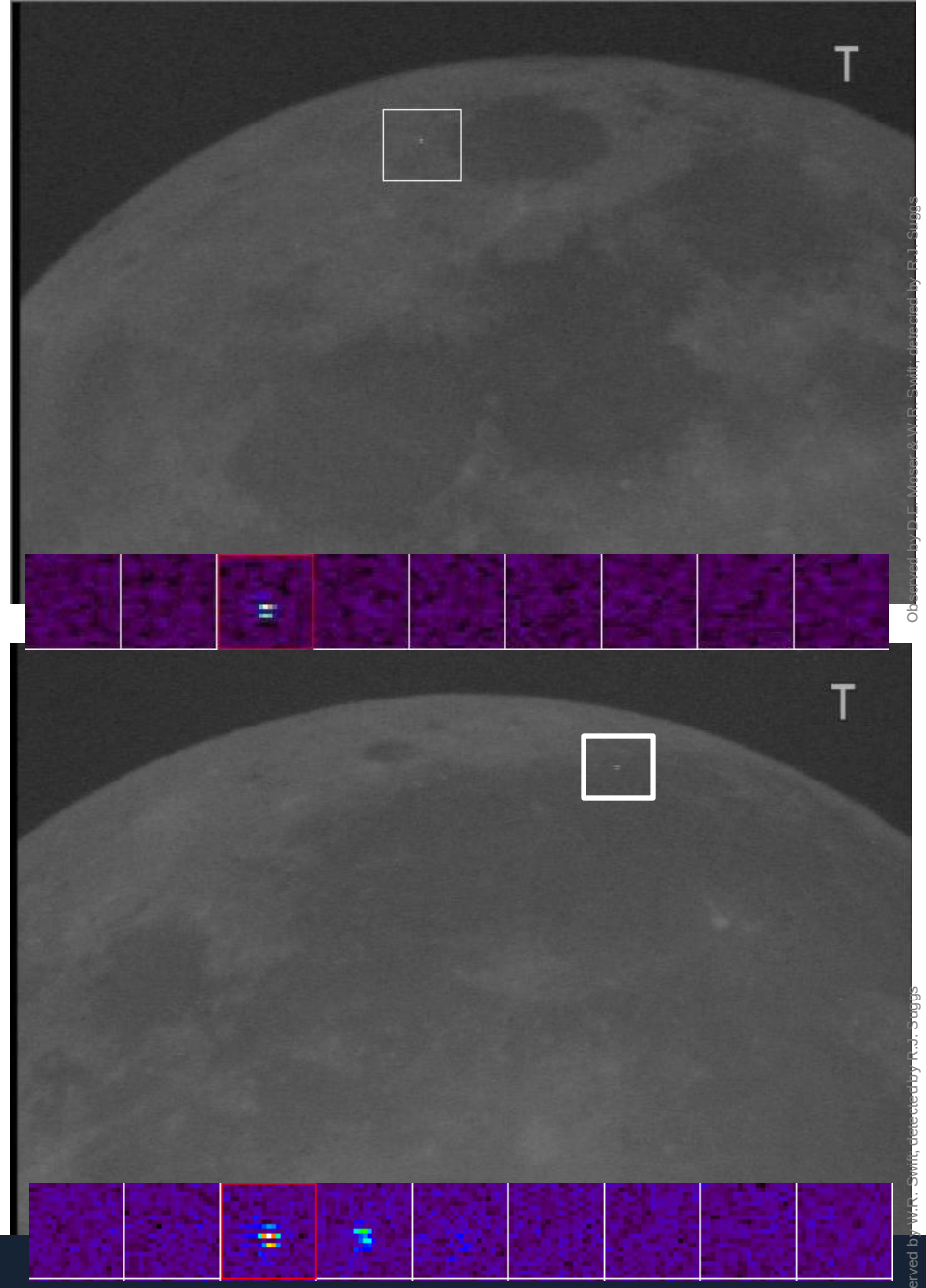
LunarScan

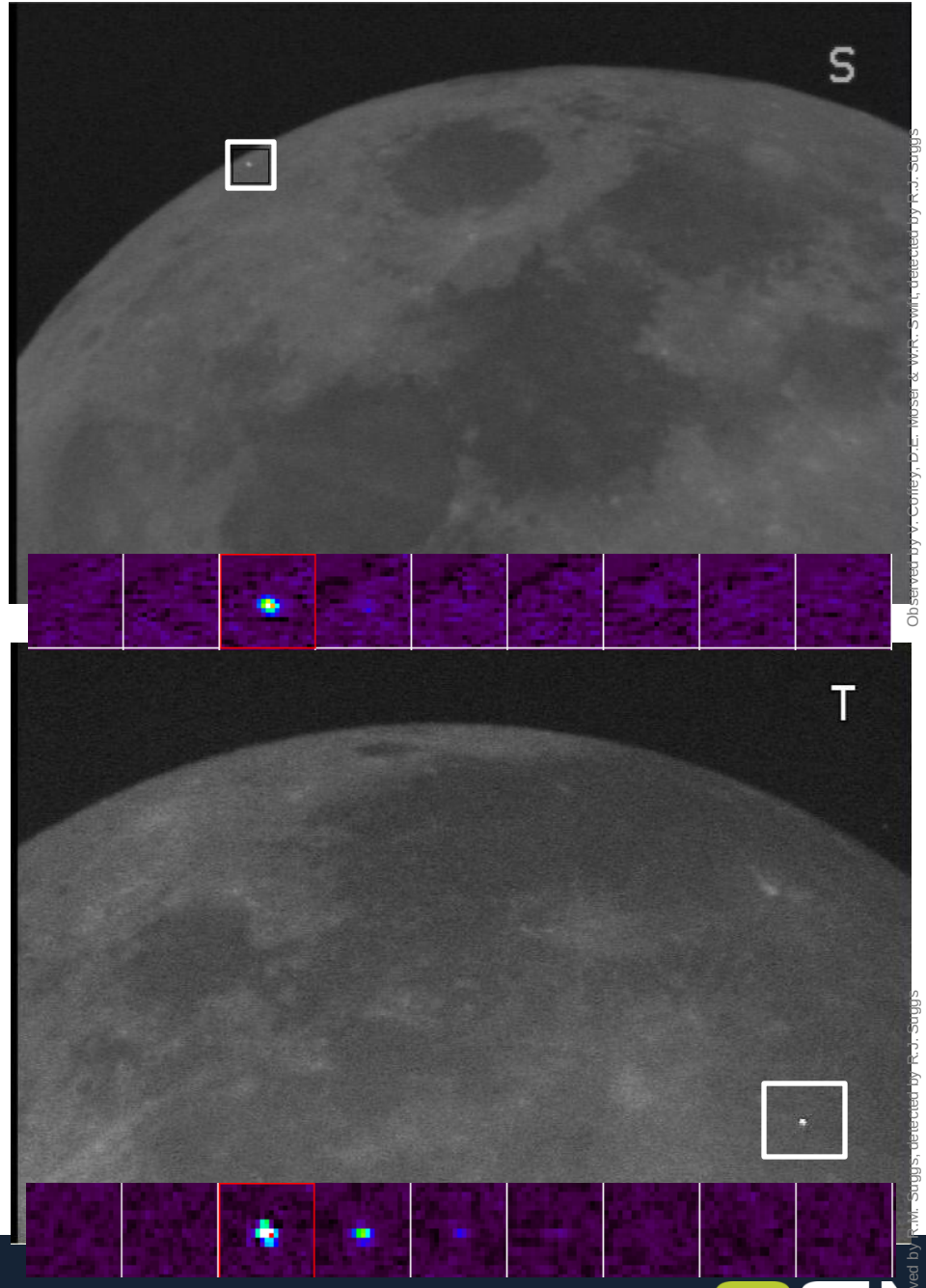


Impact 15 Dec 2006

- Field of view encompasses approximately 4 million square kilometers of lunar surface
- Resolution of ~4 km
- Complete to magnitude +9 – corresponds to a mass of about 30 g at sporadic speed of 20 km s^{-1} (ping pong ball size)
- LunarScan software (Pete Gural) used to detect impact flashes – require simultaneous detections in both telescopes
- ArcGIS software used to selenolocate impact flashes – refraction correction also applied




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Observed by NASA Swift, detected by R.J. Suggs


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Observed by A. Conley, D.E. Mosier & NASA Swift, detected by R.J. Suggs

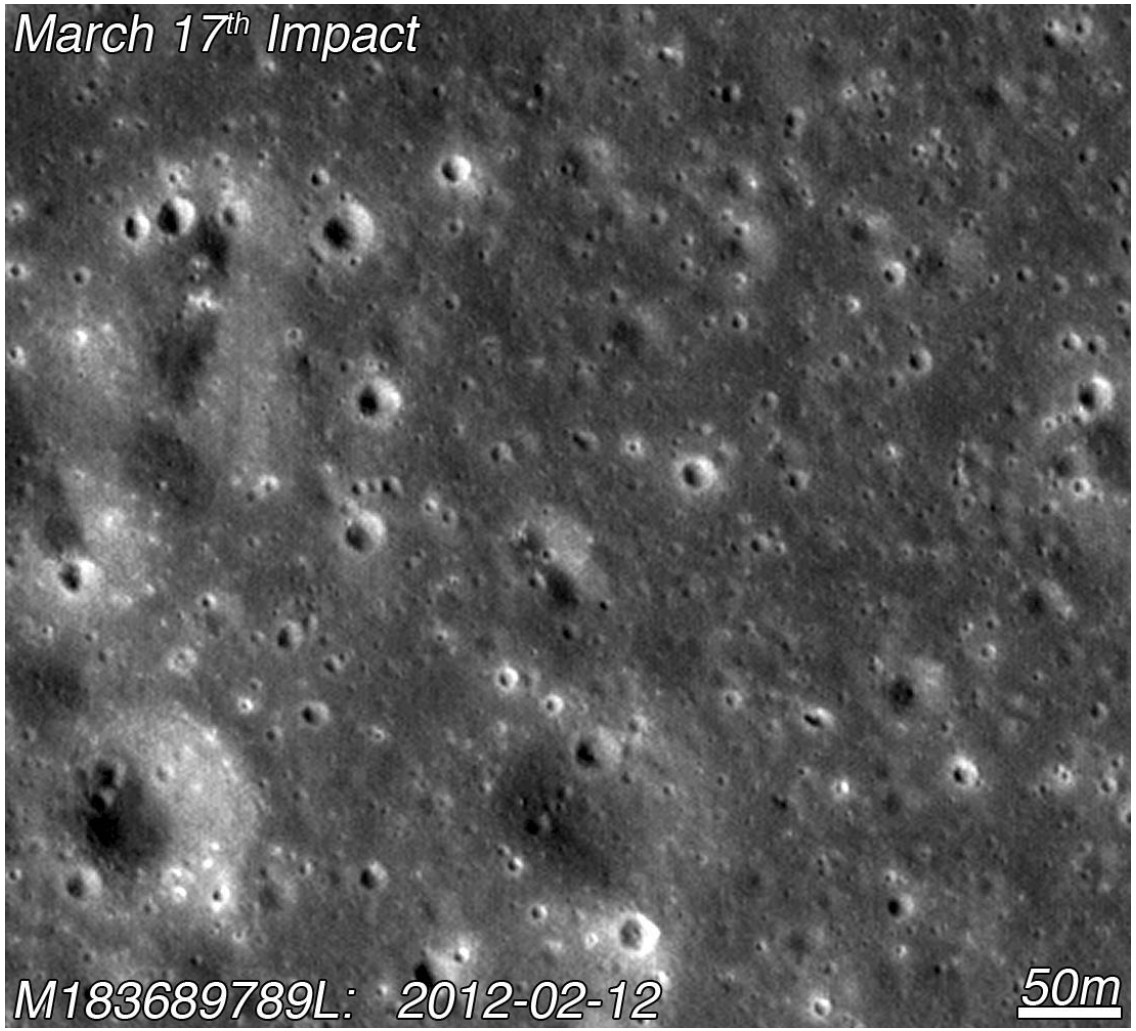
T
Observed by NASA Swift, detected by R.J. Suggs

T
Observed by NASA Swift, detected by R.J. Suggs

March 17, 2013 impact



Impact crater found by LRO



NASA LRO image provided by Robinson (2013)

Crater location

- 20.7135° N, 24.3302° W

Crater info

- Rim-to-rim diameter = 18 m
- Inner diameter = 15 m
- Depth ≈ 5 m

Features

- Fresh, bright ejecta
- Circular crater
- Asymmetrical ray pattern

Impactor mass estimated to be ~16 kg

Current status

- 443 impacts detected as of this date
- A new version of LunarScan developed in 2020 to detect flashes in real time
- Pandemic forced suspension of observations; observatory currently being refitted and brought back online
- Conducting ”practice” observations with new personnel during last quarter of this year

